

WEEK 3 REPORT

Superstore Sales and Profit Analysis Using Power BI

1. Problem Statement

Superstore sales data contains information about customer orders, products, sales, quantity, profit, regions, categories and order dates. The objective of this project is to analyze sales and profit performance using Microsoft Power BI and present the results through interactive visualizations.

The project focuses on identifying sales trends, analyzing category and regional performance, and understanding the relationship between sales and profit over different time periods.

2. Objectives

The main objectives of this project are:

- To import and prepare Superstore sales data using Power BI.
 - To clean and transform the dataset using Power Query.
 - To create a dedicated Date Table for time-based analysis.
 - To create a Year–Quarter–Month–Day date hierarchy.
 - To establish relationships between the Date Table and sales data.
 - To create DAX measures for sales and profit analysis.
 - To apply time-intelligence calculations.
 - To analyze category, region and segment performance.
 - To develop an interactive dashboard for sales and profit analysis.
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3. Dataset

The Superstore dataset contains information about customer orders, products, sales and profitability.

Column	Description
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Customer ID	Unique customer identification

Customer Name	Name of the customer
Product ID	Unique product identification
Product Name	Name of the product
Category	Product category
Sub-Category	Product sub-category
Segment	Customer segment
Region	Geographical region
Sales	Sales amount
Quantity	Number of products ordered
Profit	Profit generated from the order

The dataset was used to analyze overall business performance and identify important sales and profit trends.

4. Power Query

The Superstore sales dataset was imported into Microsoft Power BI and prepared using Power Query.

The data was checked and transformed using appropriate data types:

- Order Date → Date
- Ship Date → Date
- Customer ID → Text
- Product ID → Text
- Category → Text
- Sub-Category → Text
- Segment → Text
- Region → Text
- Sales → Decimal Number
- Quantity → Whole Number
- Profit → Decimal Number

	date	year	quarter	month	month number	day	day of week
1	2019-01-01	2019	1	1	1	1	1
2	2019-01-02	2019	1	1	2	2	2
3	2019-01-03	2019	1	1	3	3	3
4	2019-01-04	2019	1	1	4	4	4
5	2019-01-05	2019	1	1	5	5	5
6	2019-01-06	2019	1	1	6	6	6
7	2019-01-07	2019	1	1	7	7	7
8	2019-01-08	2019	1	1	8	8	8
9	2019-01-09	2019	1	1	9	9	9
10	2019-01-10	2019	1	1	10	10	10
11	2019-01-11	2019	1	1	11	11	11
12	2019-01-12	2019	1	1	12	12	12
13	2019-01-13	2019	1	1	13	13	13
14	2019-01-14	2019	1	1	14	14	14
15	2019-01-15	2019	1	1	15	15	15
16	2019-01-16	2019	1	1	16	16	16
17	2019-01-17	2019	1	1	17	17	17
18	2019-01-18	2019	1	1	18	18	18
19	2019-01-19	2019	1	1	19	19	19
20	2019-01-20	2019	1	1	20	20	20
21	2019-01-21	2019	1	1	21	21	21

Power Query was used to clean and transform the dataset before loading it into the Power BI data model.

The data was checked for missing values, incorrect data types and duplicate records to ensure that the dataset was suitable for analysis.

5. Date Table

A dedicated Date Table was created using DAX to support time-based analysis.

The Date Table contains:

- Date
- Year
- Quarter
- Month
- Month Number
- Day
- Day of Week

Date Table
Date
Year
Quarter
Month
Month Number
Day
Day of Week

The Date Table provides a structured calendar that can be used for time-based calculations and filtering.

It also helps in analyzing sales and profit across different periods such as years, quarters and months.

6. Date Hierarchy

A date hierarchy was created to enable drill-down analysis.

Hierarchy:

Year → Quarter → Month → Day



This hierarchy allows the sales data to be analyzed at different levels of time.

For example, users can start with yearly sales and drill down into quarters, months and individual days.

7. Relationship

A relationship was established between the Date Table and the Superstore Sales Data table.

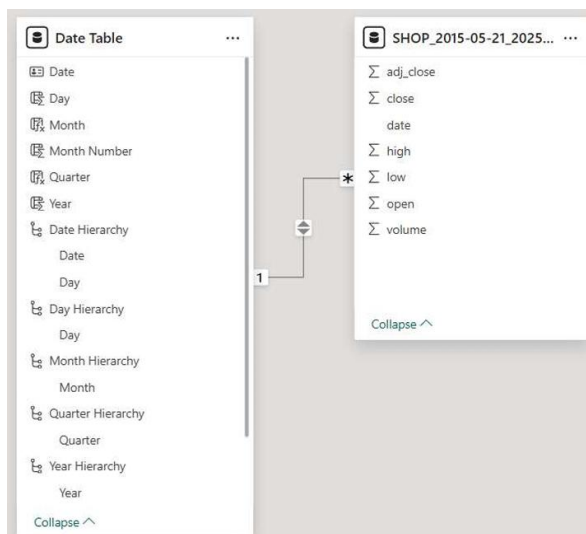
Relationship:

`Date Table[Date] → Superstore Sales[Order Date]`

Relationship Type: One-to-Many (1:*)

The Date Table acts as the **one-side**, while the Superstore Sales table acts as the **many-side**.

This relationship allows the Date Table to filter the sales data correctly and supports time-based analysis in Power BI.



8. DAX Measures

DAX measures were created to calculate important sales and business performance metrics.

Measure	Purpose
Total Sales	Calculates the total sales amount
Total Profit	Calculates the total profit generated
Total Orders	Calculates the total number of orders
Total Quantity	Calculates the total quantity of products sold
Average Sales	Calculates the average sales value
Profit Margin %	Calculates the percentage of profit compared with sales

DAX Measures and Calculations

Total Sales

```
Total Sales = SUM(Superstore[Sales])
```

Total Profit

```
Total Profit = SUM(Superstore[Profit])
```

Total Quantity

```
Total Quantity = SUM(Superstore[Quantity])
```

Total Orders

```
Total Orders = DISTINCTCOUNT(Superstore[Order ID])
```

Average Sales

```
Average Sales = AVERAGE(Superstore[Sales])
```

Profit Margin %

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

These measures help in understanding the overall sales, profitability and order performance of the Superstore dataset.

9. Time Intelligence

Time Intelligence is used to analyze business performance across different time periods.

In this project, time-intelligence calculations were used to analyze sales and profit trends over time.

Functions Used

Function	Purpose
DATEADD	Moves the date context to another time period
TOTALYTD	Calculates the value from the beginning of the year up to the selected date
SAMEPERIODLASTYEAR	Compares the selected period with the same period of the previous year

These calculations help in understanding changes in sales and profit across different periods and support year-based business analysis.

10. Dashboard

An interactive **Superstore Sales and Profit Analysis Dashboard** was created using Power BI.

KPI Cards

The dashboard contains KPI cards for:

- Total Sales
- Total Profit

- Total Orders
- Total Quantity
- Average Sales
- Profit Margin %

Slicers

The following slicers were used:

- Year
- Category
- Region
- Segment

Visualizations

The dashboard contains the following visualizations:

- **Category-wise Sales** – Clustered Column Chart
- **Region-wise Profit** – Clustered Column Chart
- **Sales Over Time** – Line Chart
- **Sales and Profit Analysis** – Interactive visuals
- **Decomposition Tree** – Total Sales by Category
- **KPI Cards** – Overall business performance

The slicers allow users to select specific years, categories, regions and customer segments and dynamically analyze the data.

11. Insights

The dashboard provides the following analytical insights:

- KPI cards provide a quick overview of sales, profit, orders and quantity.
 - Category-wise analysis helps identify the sales contribution of different product categories.
 - Region-wise analysis helps understand profit performance across geographical regions.
 - The sales trend visual shows how sales change over different time periods.
 - Segment analysis helps understand the contribution of different customer groups.
 - The decomposition tree provides a detailed breakdown of total sales by category and other dimensions.
 - The slicers allow users to interactively filter the dashboard.
 - Time-intelligence calculations support comparison of business performance across different periods.
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12. Conclusion

The Week 3 project successfully demonstrated the process of importing, cleaning and transforming Superstore sales data using Microsoft Power BI. A dedicated Date Table and date hierarchy were created to support time-based analysis.

Relationships were established between the Date Table and the sales data, and DAX measures were developed to calculate important business metrics such as Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales and Profit Margin.

Time-intelligence calculations were also applied to analyze business performance across different periods. Finally, an interactive dashboard was created using KPI cards, charts, slicers and a decomposition tree to provide a clear and structured analysis of Superstore sales and profit performance.